ARIC

مركز الابتكار والأبحاث المتقدمة

Advanced Research and Innovation Center

PARTNERS

MUBADALA

مبادلة للرعاية الصحية

Mubadala Health

STRATA ستراتا

A Mubadala Company

BOEING

Sanad

A Mubadala Company

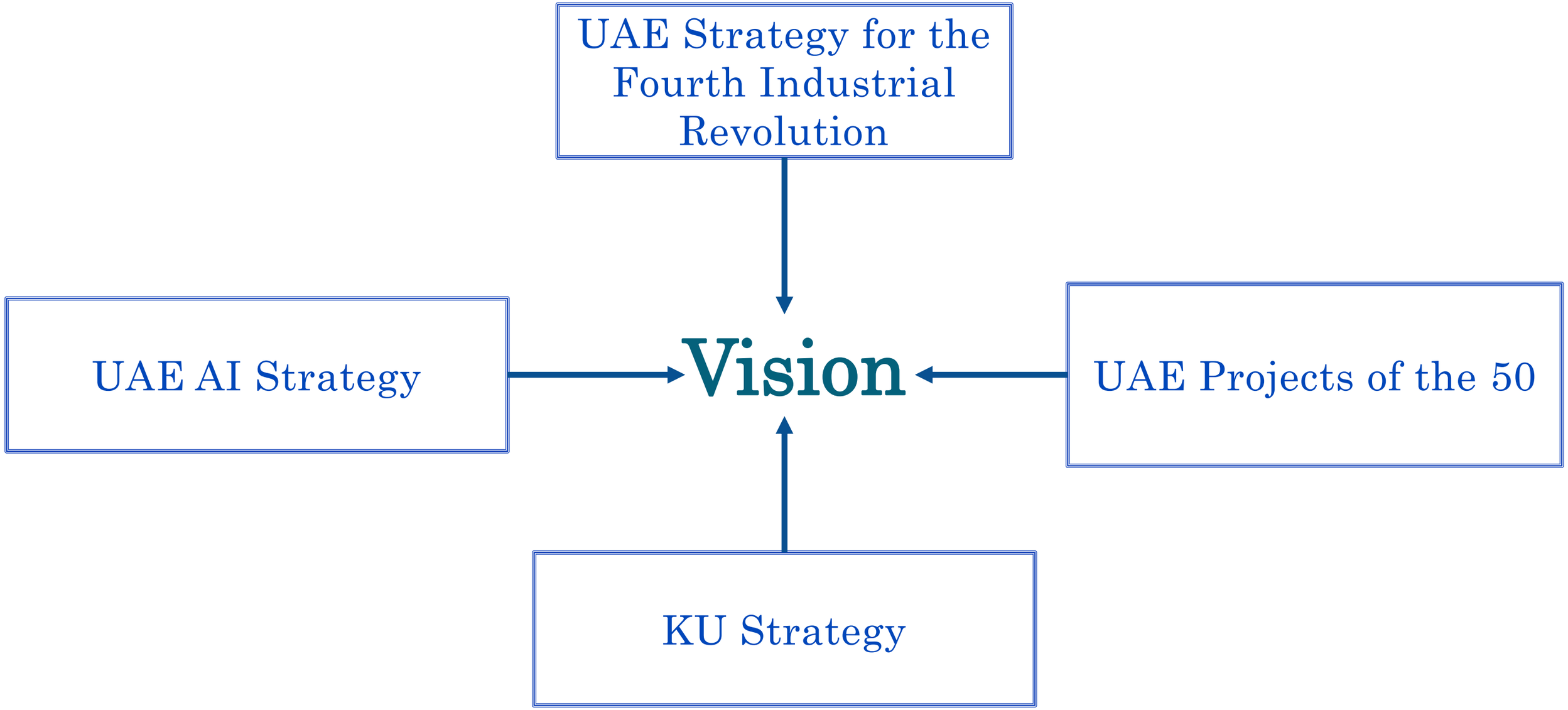
Lufthansa

جامعة خليفة

Khalifa University

صندوق الوطن

SANDOOQ AL WATAN


UAE Strategy for the Fourth Industrial Revolution
UAE AI Strategy
Vision
UAE Projects of the 50
KU Strategy

https://www.aric.ae/

Vision: Our vision is to become a regional and global leader in the research, development, demonstration and deployment of advanced manufacturing, health, and energy technologies.

Mission: Our mission is to develop research, human capital and technology innovation that underpin the UAE's manufacturing, health and energy sectors. We are committed to bridging the gap between academia and industry through the development and deployment of solutions that address real-world challenges and needs.

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Aim: To Put Innovation into Practice and Create A Competitive Edge in the Advanced Industries

Objectives:

- To identify technologies that have the potential to create and transform the manufacturing, health and energy sectors.
- To support UAE industries in their efforts to develop innovative technology solutions for manufacturing, health and energy.
- To educate and train students in the design and manufacture of advanced technologies.

Funding

(1) Aerospace Holding Company LLC (AHC), a limited liability company established under the laws of the Emirates of Abu Dhabi, and whose principal address is PO. Box 45005, Abu Dhabi , United Arab Emirates (**AHC**); and
(2) Khalifa University of Science and Technology, an independent legal entity established by Law number (5) for the year 2017, as amended, having its place of business in the UAE Emirate of Abu Dhabi, P.O. Box 127788 (**Khalifa University**).

The cash funding allocation from AHC and Khalifa University for the Center shall be paid according to the following schedule:

Year	2023	2024	2025	2026
AHC contribution (AED) excluding VAT	2,500,000	2,500,000	2,500,000	2,500,000
Khalifa University contribution (AED)	2,000,000	2,000,000	2,000,000	2,000,000
Total cash contributions (AED)	4,500,000	4,500,000	4,500,000	4,500,000

ARIC

2023 KPIs

Objective	KPI	Target	Basis
To identify technologies that have the potential to create and transform the manufacturing, health, and energy sectors.	Number of papers accepted/published by top 10% journals (per Scopus)	12	1 top 10% paper per $75k spent. Due to the involvement of ARIC in developing advanced industrial solutions, the basis is 1 paper per $75k in research spent. The 2023 KPI considers the AED3.5M budget, after deducting the AED1M deficit.
	Number of filed patents	1	One invention disclosure per $1.5M R&D spent
To support UAE industries in their efforts to develop innovative technology solutions for manufacturing, health and energy.	Number of high TRL technologies being qualified/ deployed for end-user	2	2 projects to achieve TRL 8/9
	Number of technologies being tested/evaluated for industrial application	2	2 projects to achieve TRL 6
	Commercialization: Number of Startup spin-offs or licensing agreements based on ARIC IP	1	1 spin-off or 1 licensing agreement based on BIP, every two years (per $3M R&D spent). The achievement of this KPI will be assessed at the beginning of 2025
	Communications: Number of events where ARIC technologies are showcased	3	Planned Events: • IDEX in February 2023 • Make it in the UAE, May 2023 • Dubai Airshow in November 2023 • ICRA, London 2023
	Number of articles in local or international media, and LinkedIn posts	12	One article or LinkedIn post per month throughout 2023
To educate and train students in the design and manufacture of advanced technologies.	Number of UAE nationals being deployed on ARIC projects	10	One new SDP group per semester with an average of 5 students per group

Research Themes

Advanced Materials

Robotics and Digital Systems

Health Technologies

Energy and Environmental Technologies

Research Themes

Theme I: Theme leader, Dr. Andreas Schiffer

Advanced Materials*

* Includes Additive Manufacturing, Advanced Composite Processing Techniques, Thermoplastic Materials and Processes, Graphene Applications, and Multi-Functional Materials

Theme I: Advanced Materials

Planned projects:

For STRATA:

1. **Thermoplastic and graphene as a liner for hydrogen tanks**, pressure 700 bar, a stringent requirement for allowable leakage (low permeability).
   - Faculty: Prof. Lianxi Zheng (PI) and Prof. Kin Liao (co-PI)
   - Researchers: Eng. Pradeep George, Eng. Mohammed Al Krunz

For Sanad Aerotech:

1. **Gas Turbine blade repair**, using 3D metallic printing with a proper interaction
   - Faculty: Dr. Andreas Schiffer (PI) and Dr. Kamran Khan (co-PI)
   - Researchers: New RA, (offer was sent, and the candidate accepted the offer in 30th March 2023)
   - Students: One new PhD (expected to join, June 2023)

Research Themes

Theme II: Theme leader, Dr Yahya Zweiri

Robotics and Digital Systems*

* Includes Robotic Manufacturing and Assembly Technologies, Robotic Inspection Systems, Industrial Internet of Things (IIOT), Artificial intelligence (AI), Virtual Intelligence (VI), and Blockchain

Theme II: Robotics and Digital Systems

Planned projects:

For STRATA:

1. **Qualification and Deployment of novel deburring end-effector**
2. **Development of inner-hole painting end-effector**
3. **Automated visual inspection and control for fully automated production**

For Sanad Aerotech:

1. **Automated visual inspection of blades**
2. **Laser-based coordinate measurements for MRO operations**
3. **Robotic sorting of engine components**

Theme II: Robotics and Digital Systems

Resources:

1. STRATA cluster
    - Faculty: Dr. Yahya Zweiri
    - Researchers: Abdulla Ayyad, Laith AbuAssi, and one new RA (offer was sent)
    - Students: Oussama Abdulhay (PhD), Islam Zaid (MSc student), Hussain Sajwani (NSR), and Falah Alkhoori (NSR)

2. SANAD cluster:
    - Faculty: Dr. Yahya Zweiri and Dr. Yusra Abdulrahman
    - Researchers: Mohammed Ramadan, and one new RA (offer was sent)
    - Students: Mohammed Ali (PhD)

Research Themes

Theme III: Theme leader, TBD

Health Technologies*

* Includes Digital Solutions such as Artificial Intelligence (AI), Virtual Reality (VR), and Blockchain, Robotic Solutions at Point of Care (POC), Additive Manufacturing for Medical Devices and Surgical Tools, Bio-sensing, and Exoskeleton

Theme III: Health Technologies

Planned projects:

1. **Exoskeleton project**
2. **Intelligent tactile sensing**

Theme III: Health Technologies

Resources:

1. Exoskeleton project
   - Faculty: Dr. Irfan Hussain
   - Researchers: Ahmed Al Jarrah
   - Students: Humaid Ibrahim (NSR), Ali Al Hammadi (NSR), and two new MSc students (joined on Feb 2023).

2. Intelligent tactile sensing
   - Faculty: Dr. Yahya Zweiri
   - Researchers: One new Postdoc (pending clearance) and one new RA (offer was sent),
   - Students: One new PhD (expected to join, June 2023)

Research Themes

Theme IV:

Energy and Environmental Technologies*

*Includes Water Filtration and Desalination, Energy Storage, Alternative Energy Solutions, Highly Efficient Heat Exchangers for Cooling, and Cathode and Anode Materials for Batteries and Energy Storage Systems

Theme IV: Energy and Environmental Technologies

Planned projects:

External Engagements:

2. Lightweight, sustainable and energy-efficient designs and production methods for upcycled Graphene reinforced hybrid Composites (Light-G-Comp), RIC-2D call. It is expected in June-July 2023.

Theme IV: Energy and Environmental Technologies

Resources:

Lightweight, sustainable and energy-efficient designs and production methods for upcycled Graphene reinforced hybrid Composites (Light-G-Comp), RIC-2D call:

- Faculty: Prof. Jang Kyo Kim , Dr. Ebrahim Al Hajri, and Dr. Yahya Zweiri, + External collaborators
- Researchers: One new postdoc and one new RA
- Students: One new PhD student and one new MSc student

Structure

ARIC Senior Management Board
- 3 Members from KU
- 3 Members from Mubadala / STRATA

ARIC Director

Supporting Faculty

STRATA / SANAD Personnel

Theme I: Advanced Materials
- Lab engineer
- Postdoc
- Research Associate / Engineer

Theme II: Robotics and Digital Systems
- Postdoc
- Research Associate / Engineer

Theme III: Health Technologies
- Postdoc
- Research Associate / Engineer

Theme IV: Energy and Environmental Techs.
- Postdoc
- Research Associate / Engineer

THANK YOU